Lecture 6

# Symmetry, projectors, and degeneracy

Symmetry shows, before the Hamiltonian is diagonalised, into which blocks its matrix splits and how many times, at least, its levels are degenerate. Characters give projectors that split a representation into its isotypic components. In a basis adapted to the symmetry, an operator that commutes with the group is block-diagonal, so each of its eigenvalues is at least as degenerate as the dimension of an irreducible representation. We apply this theorem to a particle hopping on a triangle, and in Lecture 7 to the vibrations of a molecule.

By the end of this lecture you will be able to:

- Say when a group is a symmetry of a quantum system.
- Project a representation onto its isotypic components, and build a symmetry-adapted basis.
- Reduce an invariant operator to its matrices $F^{(i)}$, and predict its degeneracies.

## 6.1 Symmetry in quantum mechanics

The state space of a quantum system is a complex vector space with an inner product, and a symmetry acts on it by a unitary representation. For a particle in space the states are wave functions, and the group acts by the function-space construction of Lecture 3,

$$\big(\rho(g)\psi\big)(\vec{x}\,) = \psi\big(g^{-1}\vec{x}\,\big),$$

which is unitary for $g \in \mathrm{O}(3)$: since $|\det g| = 1$, the substitution $\vec{x} \mapsto g^{-1}\vec{x}$ preserves $\int |\psi|^2$. Physical models often restrict attention to a few states. The state space is then finite-dimensional; Example 6.3 below is one.

**Definition 6.1** (Quantum symmetry)**.** Let $\rho$ be a unitary representation of a group $G$ on the state space. Then $G$ is a *symmetry* of the quantum system with Hamiltonian $H$ if

$$\big[\rho(g),\, H\big] = 0 \qquad \text{for all } g \in G.$$

In other words, $H$ is an intertwiner of the representation $\rho$ with itself (Lecture 3).

**Proposition 6.2.** If $[\rho(g), H] = 0$ for all $g$, then every energy eigenspace $\{\psi : H\psi = \lambda\psi\}$ is an invariant subspace. So each eigenspace carries a representation of $G$.

*Proof.* $H\rho(g)\psi = \rho(g)H\psi = \lambda\,\rho(g)\psi$. ∎

So the energy levels of a symmetric system are representations of its symmetry group. Which representations, and how large? Answering that is the business of this lecture. The answer needs only linearity: we shall apply the same theorem to the small-vibration model of Lecture 7.

**Example 6.3** (A particle hopping on a triangle)**.** Put one electron orbital on each of three identical atoms at the corners of an equilateral triangle — the three hydrogens of ammonia, or the three carbons of the cyclopropenyl ring — and write $|j\rangle$ for the orbital on atom $j$ (Figure 6.1). An orbital is a wave function centred on its atom, so a symmetry of the triangle carries the orbital on atom $j$ to the orbital on the atom where $j$ goes: $\rho(g)\,|j\rangle = |g(j)\rangle$, the permutation representation of Lecture 3. Keeping only these three orbitals, taken orthonormal, $\langle j|k\rangle = \delta_{jk}$, the state space is $\mathbb{C}^3$. The electron hops between neighbouring atoms with amplitude $-t$, where $t > 0$, and on a triangle every two atoms are neighbours, so

$$H = -t \sum_{j \neq k} |j\rangle\langle k| = -t\,(J - \mathbb{1}) = -t \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix},$$

in the basis $|1\rangle, |2\rangle, |3\rangle$, with $J$ the $3 \times 3$ matrix of ones. The matrices of $C_{3v} \cong S_3$ are permutation matrices, and $H$ commutes with every one of them, because each commutes with $J$. So $C_{3v}$ is a symmetry. The levels turn out to be $-2t$ and a doubly degenerate $t$ (Figure 6.1(b)). Why is $t$ doubled, and can a symmetric change of $H$ split it? The rest of the lecture answers both questions, and finds the eigenvectors without diagonalising $H$ by hand. ◁

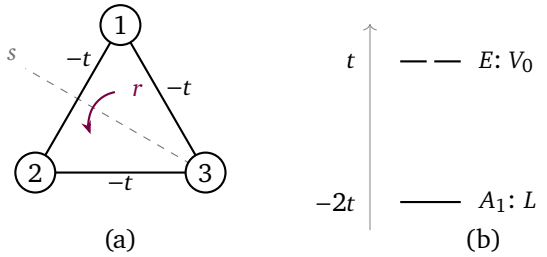



Figure 6.1: The particle on a triangle. (a) Orbitals $|j\rangle$, hopping $-t$, the rotation $r$ and the mirror $s$ (dashed). (b) The levels for $t > 0$: the singlet $A_1$ on $L$ and the doublet $E$ on $V_0$ (Section 6.4).

## 6.2 Projection operators

Characters compute not only multiplicities: together with the operators $\rho(g)$, they also find invariant subspaces. From here on $G$ is finite and $V$ is a finite-dimensional complex representation, and we keep two notations apart: superscripts in parentheses, $V^{(i)}$, denote the abstract irreducible representations, and subscripts, $V_i$, the pieces of a given representation $V$ defined next.

**Definition 6.4** (Isotypic components)**.** Let $(\rho, V)$ be a representation. The *isotypic com-*

*ponent* $V_i$ of type $i$ is the sum of all irreducible invariant subspaces of $V$ equivalent to $V^{(i)}$.

If $V \cong \bigoplus_i m_i V^{(i)}$, the component $V_i$ collects the $m_i$ copies of $V^{(i)}$, so $\dim V_i = m_i n_i$ with $n_i = \dim V^{(i)}$. When $m_i > 1$, the individual copies inside $V_i$ are *not* unique, but the component is: the next theorem produces it from the character and the operators $\rho(g)$, without choosing a decomposition.

**Theorem 6.5** (Character projectors)**.** For each irreducible representation define

$$P_i = \frac{n_i}{|G|} \sum_{g \in G} \overline{\chi_i(g)}\, \rho(g).$$

Then $P_i$ commutes with every $\rho(h)$, and it acts as the identity on every irreducible invariant subspace of type $i$ and as zero on those of every other type. So $P_i$ is the projection onto $V_i$,

$$\sum_i P_i = \mathrm{id}_V \quad \text{and} \quad V = \bigoplus_i V_i.$$

If $\rho$ is unitary, every $P_i$ is Hermitian, and the components are mutually orthogonal.

*Proof.* First, conjugate $P_i$ by $\rho(h)$ and rename $g' = hgh^{-1}$:

$$\rho(h) P_i \rho(h)^{-1} = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)}\, \rho(hgh^{-1}) = \frac{n_i}{|G|} \sum_{g'} \overline{\chi_i(h^{-1}g'h)}\, \rho(g') = P_i,$$

because $\chi_i$ is a class function. Next, let $W$ be an irreducible invariant subspace of type $j$. The operator $P_i$ maps $W$ into itself and commutes with the action on it, so by Schur's lemma $P_i|_W = \kappa\, \mathrm{id}_W$. The trace of this equality fixes $\kappa$:

$$\kappa n_j = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)}\, \chi_j(g) = n_i \langle \chi_i, \chi_j \rangle_G = n_i \delta_{ij}.$$

So $P_i|_W = \mathrm{id}_W$ when $W$ is of type $i$, and $P_i|_W = 0$ otherwise. By Maschke's theorem $V = W_1 \oplus \cdots \oplus W_r$ with every $W_k$ an irreducible invariant subspace. Each $W_k$ has exactly one type, so $\sum_i P_i$ acts as the identity on every $W_k$, that is $\sum_i P_i = \mathrm{id}_V$, and $P_i$ is the projection onto the sum of the $W_k$ of type $i$, with kernel the sum of the others. That sum is all of $V_i$, because every $W$ of type $i$ has $W = P_i W \subseteq \operatorname{im} P_i$, so it does not depend on the decomposition. In particular $V_j \subseteq \ker P_i$ for $j \neq i$. Finally, if $\rho$ is unitary, then $\rho(g)^\dagger = \rho(g^{-1})$ and $\overline{\chi_i(g)} = \chi_i(g^{-1})$, so renaming $g \to g^{-1}$ in the sum gives $P_i^\dagger = P_i$. Then $\langle P_i v, w \rangle = \langle v, P_i w \rangle = 0$ for every $w \in V_j$, $j \neq i$, so $V_i \perp V_j$. ∎

**Example 6.6** (The projectors on the triangle's orbitals)**.** Recall from Lecture 2 that $C_{3v} \cong S_3$ has the six elements $e, r, r^2, s, rs, r^2 s$, with $r$ the rotation by $2\pi/3$ and $s$ a reflection. On the orbitals of Example 6.3 they act by permutation matrices: $r$ by the 3-cycle $(1\,2\,3)$ and $s$ by the transposition $(1\,2)$, the mirror of Figure 6.1,

$$D(r) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad D(s) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and the other four matrices are their products. For each $j$, the three elements $e, r, r^2$ send it to 1, 2 and 3, once each, and so do the three elements $s, rs, r^2 s$. So every column of each of

the two sums below is $(1,1,1)$, and with $J$ the all-ones matrix,

$$D(e)+D(r)+D(r^2)=J=D(s)+D(rs)+D(r^2s).$$

The characters are real and constant on classes. From Lecture 5,

| | $\{e\}$ | $\{r,r^2\}$ | $\{s,rs,r^2s\}$ |
|---|---|---|---|
| $\chi_{A_1}$ | 1 | 1 | 1 |
| $\chi_{A_2}$ | 1 | 1 | $-1$ |
| $\chi_E$ | 2 | $-1$ | 0 |

Applying the formula of Theorem 6.5 gives $P_{A_2}=\frac{1}{6}(J-J)=0$ and

$$P_{A_1}=\frac{1}{6}(J+J)=\frac{1}{3}\begin{pmatrix}1&1&1\\1&1&1\\1&1&1\end{pmatrix},\qquad P_E=\frac{2}{6}\Big(2\,\mathbb{1}-(J-\mathbb{1})\Big)=\frac{1}{3}\begin{pmatrix}2&-1&-1\\-1&2&-1\\-1&-1&2\end{pmatrix},$$

since $D(r)+D(r^2)=J-\mathbb{1}$. Here $P_{A_2}=0$ records that $A_2$ does not occur in the decomposition. The image of a matrix is spanned by its columns. Every column of $P_{A_1}$ is a multiple of $u_0=(1,1,1)$, and every column of $P_E$ has components summing to zero. With the dimensions given by the traces,

$$\operatorname{im}P_{A_1}=L=\mathbb{C}u_0,\quad \dim L=\operatorname{tr}P_{A_1}=1,\qquad \operatorname{im}P_E=V_0,\quad \dim V_0=\operatorname{tr}P_E=2,$$

where $V_0$ is the plane made of the vectors whose components sum to zero. So the projectors have produced, without any guessing, the decomposition $\mathbb{C}^3=L\oplus V_0$ of Lecture 4, and $P_{A_1}$ is the averaging operator of Lecture 5. The dimensions 1 and 2 are those of the two levels of Example 6.3, the singlet and the doublet of Figure 6.1(b); Section 6.4 shows that $H$ is a scalar operator on each. ◁

## 6.3 The block-diagonalisation theorem

Let $A\in\operatorname{Hom}_G(V,V)$, that is, a linear operator on $V$ that commutes with every $\rho(g)$. Physically, $A$ is a Hamiltonian, or the dynamical matrix of Lecture 7. Since each $P_i$ is a combination of the $\rho(g)$,

$$A\rho(g)=\rho(g)A\ \text{ for all } g\quad\Longrightarrow\quad AP_i=P_iA\quad\Longrightarrow\quad AV_i\subseteq V_i.$$

So $A$ has no matrix elements between different components. What does it do inside one?

Choose a *symmetry-adapted basis* of $V_i\cong m_i\,V^{(i)}$,

$$w_{\nu,a},\qquad \nu=1,\dots,n_i,\quad a=1,\dots,m_i.$$

Here $\nu$ labels the basis vectors of each copy of $V^{(i)}$, called its *partners*, and $a$ counts the copies (Figure 6.2). For every $a$ the vectors $w_{\nu,a}$ transform by the same matrices $D^{(i)}(g)$, the matrices of $V^{(i)}$ in a basis fixed once; such copies are called *aligned*:

$$\rho(g)\,w_{\nu,a}=\sum_{\mu}D^{(i)}_{\mu\nu}(g)\,w_{\mu,a}.$$

Such a basis exists, and with an invariant inner product it can be chosen orthonormal. The $D^{(i)}(g)$ are then unitary (Section 6.5).

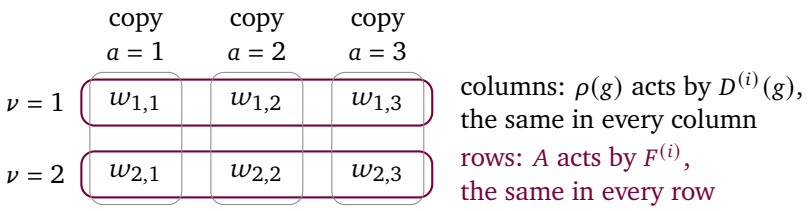



Figure 6.2: A symmetry-adapted basis of $V_i$ for $n_i = 2$ partners (rows) and $m_i = 3$ copies (columns).

The Kronecker product puts the transformation law in a compact form. Recall that for matrices $X, X', Y, Y'$ and columns $x, x', y, y'$ of matching sizes,

$$(X \otimes Y)(X' \otimes Y') = XX' \otimes YY', \quad (X \otimes Y)(x \otimes y) = Xx \otimes Yy, \quad \langle x \otimes y, x' \otimes y' \rangle = \langle x, x' \rangle \langle y, y' \rangle .$$

List the adapted basis partner by partner, along the rows of Figure 6.2, so that $w_{\nu,a}$ has the coordinates $e_\nu \otimes e_a$, with $e_\nu$ and $e_a$ the standard basis vectors of $\mathbb{C}^{n_i}$ and $\mathbb{C}^{m_i}$. Then

$$\big(D^{(i)}(g) \otimes \mathbb{1}_{m_i}\big)(e_\nu \otimes e_a) = D^{(i)}(g)\, e_\nu \otimes e_a = \sum_\mu D^{(i)}_{\mu\nu}(g)\, e_\mu \otimes e_a,$$

which is the transformation law: $\rho(g)$ acts on $V_i$ by $D^{(i)}(g) \otimes \mathbb{1}_{m_i}$.

**Theorem 6.7** (Block diagonalisation)**.** Let $A$ commute with every $\rho(g)$, and take a symmetry-adapted basis of $V_i$. There is an $m_i \times m_i$ matrix $F^{(i)}$, the *reduced matrix*, such that

$$A\, w_{\nu,b} = \sum_{a=1}^{m_i} F^{(i)}_{ab}\, w_{\nu,a} \qquad \text{for every partner } \nu.$$

In the coordinates $e_\nu \otimes e_a$, this says that $A$ acts on $V_i$ by the block-diagonal matrix

$$\mathbb{1}_{n_i} \otimes F^{(i)} = \begin{pmatrix} F^{(i)} & & \\ & \ddots & \\ & & F^{(i)} \end{pmatrix} \qquad (n_i \text{ blocks}).$$

Conversely, every such matrix commutes with every $D^{(i)}(g) \otimes \mathbb{1}_{m_i}$. In an orthonormal basis,

$$F^{(i)}_{ab} = \big\langle w_{\nu,a}, A\, w_{\nu,b} \big\rangle \quad \text{for any } \nu,$$

and $F^{(i)}$ is Hermitian when $A$ is. If $F^{(i)} v = \lambda v$ with $v \neq 0$, the $n_i$ vectors $\psi_\nu = \sum_a v_a\, w_{\nu,a}$ are independent eigenvectors of $A$ with eigenvalue $\lambda$.

*Proof.* In the coordinates $e_\nu \otimes e_a$, the component $V_i$ is $\mathbb{C}^{n_i} \otimes \mathbb{C}^{m_i}$, and $\rho(g)$ acts on it by $D^{(i)}(g) \otimes \mathbb{1}_{m_i}$. By the mixed-product rule,

$$\big(D^{(i)}(g) \otimes \mathbb{1}\big)(\mathbb{1} \otimes F) = D^{(i)}(g) \otimes F = (\mathbb{1} \otimes F)\big(D^{(i)}(g) \otimes \mathbb{1}\big),$$

so every matrix $\mathbb{1} \otimes F$, with $F$ any $m_i \times m_i$ matrix, commutes with the group; this is the converse. Such matrices form a space of dimension $m_i^2$. On the other hand, $\chi_{V_i} = m_i \chi_i$, so by Lecture 5's theorem $\langle \chi_V, \chi_W \rangle_G = \dim \mathrm{Hom}_G(V, W)$ the operators on $V_i$ that commute with the group form a space of dimension

$$\dim \mathrm{Hom}_G(V_i, V_i) = \langle m_i \chi_i, m_i \chi_i \rangle_G = m_i^2.$$

So there are no others. Since $A$ preserves $V_i$ and commutes with the group, $A = \mathbb{1} \otimes F^{(i)}$ for some matrix $F^{(i)}$, that is, $A(e_\nu \otimes e_b) = e_\nu \otimes F^{(i)} e_b$, which is the displayed formula. In an orthonormal basis the coordinates are orthonormal, so

$$\left\langle w_{\nu,a}, A\, w_{\nu,b} \right\rangle = \langle e_\nu, e_\nu \rangle \left\langle e_a, F^{(i)} e_b \right\rangle = F^{(i)}_{ab},$$

and $A^\dagger = A$ makes $F^{(i)}$ Hermitian. Finally, the coordinates of $\psi_\nu$ are $e_\nu \otimes v$, and

$$(\mathbb{1} \otimes F^{(i)})(e_\nu \otimes v) = e_\nu \otimes F^{(i)} v = \lambda\, e_\nu \otimes v.$$

The vectors $e_\nu \otimes v$ are independent because $v \neq 0$. ∎

*Group theory fixes the blocks — their sizes $m_i$, the multiplet sizes $n_i$, the adapted basis — and the dynamics only fills in the small matrices $F^{(i)}$*, which the symmetry leaves free. An irreducible that occurs once needs no diagonalisation at all: its $F^{(i)}$ is a number, and its adapted vectors are eigenvectors.

**Corollary 6.8** (Symmetry-enforced degeneracy)**.** For every number $\lambda$,

$$\dim \ker(A - \lambda\, \mathrm{id}_V) = \sum_i n_i \, \dim \ker\!\left(F^{(i)} - \lambda \mathbb{1}_{m_i}\right).$$

So an eigenvalue with an eigenvector in $V_i$ is at least $n_i$-fold degenerate. If the eigenvalues of each reduced matrix are simple and no two reduced matrices share one, each eigenspace of $A$ is exactly one irreducible multiplet.

*Proof.* Since $A$ preserves every $V_i$ and $V = \bigoplus_i V_i$, the eigenspace of $A$ is the direct sum of its parts in the $V_i$. By Theorem 6.7, on $V_i$ the matrix of $A - \lambda\, \mathrm{id}_V$ is $\mathbb{1}_{n_i} \otimes \left(F^{(i)} - \lambda \mathbb{1}_{m_i}\right)$, so that part is $\mathbb{C}^{n_i} \otimes \ker\!\left(F^{(i)} - \lambda \mathbb{1}_{m_i}\right)$, of dimension $n_i \dim \ker(F^{(i)} - \lambda \mathbb{1}_{m_i})$. ∎

Lecture 4 promised to tell a degeneracy that the symmetry enforces from an accidental one. Now we can. A degeneracy is

- *enforced* if it is $n_i$: every operator that commutes with $G$ has it;
- *accidental* if it is larger because an eigenvalue repeats inside one $F^{(i)}$, or two reduced matrices share one, and no further symmetry explains it: a generic small perturbation that keeps all the symmetry removes it;
- *hidden* if it is larger but enforced by a symmetry not yet taken into account: a larger group, or the reality of $H$ (Section 6.5).

## 6.4 The triangle, and two stacked triangles

Back to the triangle of Example 6.3. By Example 6.6, its isotypic components are $V_{A_1} = L$ and $V_E = V_0$, each type occurring once, so both reduced matrices are numbers, which Theorem 6.7 computes in an orthonormal symmetry-adapted basis. Permutation matrices are orthogonal, so the standard inner product is invariant, and $L \perp V_0$ by Theorem 6.5. With each type occurring once there is nothing to align: any unit vector of $L$ and any orthonormal basis of $V_0$ will do, the matrices of $\rho(g)$ on $V_0$ in that basis serving as $D^{(E)}(g)$. Normalising $u_0$ and applying Gram–Schmidt to the first two columns of $P_E$ gives

$$w_1 = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}, \qquad w_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}2\\-1\\-1\end{pmatrix}, \qquad w_3 = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\end{pmatrix}.$$

Here $w_1$ spans $L$, of type $A_1$, and $w_2, w_3$ span $V_0$: they are the two partners of $E$. Since each is a unit vector, $\langle w_k, Hw_k\rangle = -t\big(\langle w_k, Jw_k\rangle - 1\big)$. Every row of $J$ sums to 3, so $Jw_1 = 3w_1$, and the components of $w_2$ and $w_3$ sum to zero, so $Jw_2 = Jw_3 = 0$. The formula for $F^{(i)}_{ab}$ in Theorem 6.7 then gives

$$F^{(A_1)} = \langle w_1, Hw_1\rangle = -t\,(3-1) = -2t, \qquad F^{(E)} = \langle w_k, Hw_k\rangle = -t\,(0-1) = t,$$

the same for both partners $k = 2, 3$. So, by Theorem 6.7, in the orthonormal basis $w_1, w_2, w_3$ the Hamiltonian is

$$\begin{pmatrix} -2t & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & t \end{pmatrix},$$

and the three vectors are its eigenvectors. This answers the question of Example 6.3: the level $t$ is doubled because $E$ is two-dimensional, and by Corollary 6.8 every Hamiltonian that commutes with $C_{3v}$ keeps it doubled.

*Two stacked triangles.* A genuine matrix $F^{(i)}$ appears as soon as an irreducible occurs twice. So put a second triangle, with orbitals $|1'\rangle, |2'\rangle, |3'\rangle$, directly above the first (Figure 6.3). The electron hops with amplitude $-t$ inside each layer, as before, and with amplitude $-t'$, where $t' > 0$, between an atom and the one directly above it, and a field across the layers raises the energy of every upper orbital by $\varepsilon \geq 0$. The six orbitals are again orthonormal, and

$$H = -t \sum_{j\neq k} \big(|j\rangle\langle k| + |j'\rangle\langle k'|\big) - t' \sum_j \big(|j'\rangle\langle j| + |j\rangle\langle j'|\big) + \varepsilon \sum_j |j'\rangle\langle j'|\,.$$

With $H_0 = -t\,(J - \mathbb{1})$ the Hamiltonian of one triangle, the matrix of $H$ in the basis $|1\rangle, |1'\rangle, |2\rangle, |2'\rangle, |3\rangle, |3'\rangle$, listed pair by pair, each atom followed by the one above it, is

$$H = \begin{pmatrix} B & -t\,\mathbb{1} & -t\,\mathbb{1} \\ -t\,\mathbb{1} & B & -t\,\mathbb{1} \\ -t\,\mathbb{1} & -t\,\mathbb{1} & B \end{pmatrix} = H_0 \otimes \mathbb{1}_2 + \mathbb{1}_3 \otimes B, \qquad B = \begin{pmatrix} 0 & -t' \\ -t' & \varepsilon \end{pmatrix}.$$

The group $C_{3v}$ turns and reflects both layers together, by the matrices $D(g) \otimes \mathbb{1}_2$, and by the mixed-product rule $H$ commutes with them, because $D(g)$ commutes with $H_0$. For $\varepsilon \neq 0$ the two layers are not alike, and no symmetry swaps them. Each layer carries a copy of $V^{(A_1)} \oplus V^{(E)}$, so

$$\mathbb{C}^6 \cong 2V^{(A_1)} \oplus 2V^{(E)} \quad \text{and} \quad m_{A_1} = m_E = 2.$$

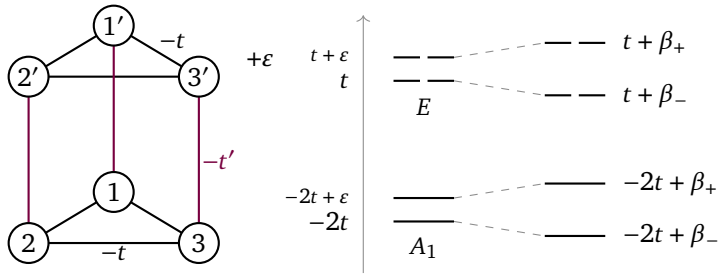
Now the isotypic component $V_{A_1}$ consists of the vectors $u_0 \otimes x$ with $x \in \mathbb{C}^2$, and $V_E$ is spanned by the vectors $y \otimes x$ with $y \in V_0$.

The copies must be aligned, and here they are built alike. In general, if the group permutes two sets of basis vectors in the same way, the map that sends each vector of the first set to its counterpart in the second is an isomorphism of representations, so the same linear combination, taken in each set, gives vectors that transform by the same matrices; Lecture 7 builds its adapted bases this way.

Here the two sets are the two layers, and each layer is one copy. Taking the vectors $w_1, w_2, w_3$ of one triangle in each layer gives the prism's adapted basis

$$w_{k,a} = w_k \otimes e_a, \qquad k = 1, 2, 3, \quad a = 1, 2,$$

with the copy $a = 1$ the lower layer and $a = 2$ the upper one; the $w_{1,a}$ are of type $A_1$, and $w_{2,a}, w_{3,a}$ are the two partners of $E$. The two copies transform by the same matrices, because




(a) (b)

Figure 6.3: Two stacked triangles. (a) Orbitals $|j\rangle$ and $|j'\rangle$, hopping $-t$ and $-t'$, the upper layer raised by $\varepsilon$. (b) The levels at $t' = 0$ (left) and $t' = t/2$ (right), for $\varepsilon = t/2$.

$(D(g) \otimes \mathbb{1}_2)(w_k \otimes e_a) = D(g)w_k \otimes e_a$. Write $h_{A_1} = -2t$ and $h_E = t$ for the reduced numbers of one triangle. By the recipe of Theorem 6.7 and the rules for $\otimes$, for $k$ of type $i$,

$$F^{(i)}_{ab} = \langle w_{k,a}, H\, w_{k,b} \rangle = \delta_{ab} \langle w_k, H_0 w_k \rangle + B_{ab} \langle w_k, w_k \rangle = h_i\, \delta_{ab} + B_{ab}.$$

So in the basis listed partner by partner, as in Theorem 6.7,

$$w_{1,1},\ w_{1,2},\ w_{2,1},\ w_{2,2},\ w_{3,1},\ w_{3,2},$$

the Hamiltonian is block diagonal:

$$\begin{pmatrix} F^{(A_1)} & 0 & 0 \\ 0 & F^{(E)} & 0 \\ 0 & 0 & F^{(E)} \end{pmatrix}, \qquad F^{(A_1)} = \begin{pmatrix} -2t & -t' \\ -t' & -2t+\varepsilon \end{pmatrix}, \qquad F^{(E)} = \begin{pmatrix} t & -t' \\ -t' & t+\varepsilon \end{pmatrix},$$

with one block $F^{(E)}$ for each partner, as the theorem promises.

Symmetry has done its part, and the dynamics is left in the $2 \times 2$ blocks. Both have the form $F^{(i)} = h_i\, \mathbb{1} + B$, so both have the orthonormal eigenvectors $v_\pm$ of $B$, with

$$B\, v_\pm = \beta_\pm\, v_\pm, \qquad \beta_\pm = \frac{\varepsilon}{2} \pm \Delta, \qquad \Delta = \sqrt{\frac{\varepsilon^2}{4} + t'^2}.$$

By the last statement of Theorem 6.7, the six vectors $w_k \otimes v_\pm$ are eigenvectors of $H$, with eigenvalues $h_i + \beta_\pm$ for $k$ of type $i$.

To sum up:

- The $6 \times 6$ problem has become two $2 \times 2$ problems, the one for $E$ solved once for both partners: two singlets $-2t + \beta_\pm$ and two doublets $t + \beta_\pm$, and $1 + 1 + 2 + 2 = 6$.
- Each doublet is enforced: for each sign, the two partners $w_2 \otimes v_\pm$ and $w_3 \otimes v_\pm$ share one eigenvalue.
- The singlet $-2t + \beta_+$ and the doublet $t + \beta_-$ meet when $\beta_+ - \beta_- = 2\Delta = 3t$, that is, for $\varepsilon < 3t$, at $t' = \frac{1}{2}\sqrt{9t^2 - \varepsilon^2}$: two reduced matrices share an eigenvalue, an accidental degeneracy.

When $\varepsilon = 0$, the layers are alike, and the symmetry grows to $C_2 \times C_{3v}$: the group $C_2$ consists of the identity and the mirror halfway between the layers, with matrix $\mathbb{1}_3 \otimes \sigma_x$ for $\sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Then $B = -t'\sigma_x$, with eigenvectors $v_- = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$ (the layers in phase, $\beta_- = -t'$) and $v_+ = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$ (in opposite phase, $\beta_+ = t'$). The vectors $w_k \otimes v_-$ are even under the mirror, and the $w_k \otimes v_+$ odd. So each isotypic component of $C_{3v}$ splits into two parts, and each is an irreducible representation of $C_2 \times C_{3v}$ that occurs once: the multiplicity $m_i = 2$ is lost, and no diagonalisation is left.

## 6.5 Degeneracy, further *[self-study]*

*Schrödinger operators.* For an unbounded $H$ the commutator needs a domain. We assume that every $\rho(g)$ maps the domain of $H$ onto itself and that $H\rho(g)\psi = \rho(g)H\psi$ there. For a Schrödinger operator this asks more than an invariant potential: boundary conditions, if any, must be invariant too. The proposition below checks the commutator on smooth functions that vanish outside some ball, a domain that every $\rho(g)$ preserves; that it then holds on the whole domain of $H$ we take for granted.

**Proposition 6.9.** Let $H = -\frac{\hbar^2}{2m}\nabla^2 + U(\vec{x}\,)$, with the group acting by the function-space action. Then $G \subset \mathrm{O}(3)$ is a symmetry if and only if the potential is invariant: $U(g\vec{x}\,) = U(\vec{x}\,)$ for all $g \in G$.

*Proof.* The Laplacian commutes with rotations and reflections. By the chain rule,

$$\partial_\alpha\big[\psi(g^{-1}\vec{x})\big] = \sum_{\beta=1}^{3} (g^{-1})_{\beta\alpha}\,(\partial_\beta\psi)(g^{-1}\vec{x}).$$

Summing the second derivatives gives the coefficients

$$\sum_{\alpha} (g^{-1})_{\beta\alpha}(g^{-1})_{\gamma\alpha} = \big(g^{-1}(g^{-1})^{\mathsf{T}}\big)_{\beta\gamma} = \delta_{\beta\gamma}$$

by orthogonality. Hence $\nabla^2\rho(g) = \rho(g)\nabla^2$. The potential term acts by multiplication, and $\rho(g)\,U\,\rho(g)^{-1}$ is multiplication by $U(g^{-1}\vec{x}\,)$. This equals multiplication by $U$ for all $g$ exactly when $U$ is invariant. ∎

*The lower bound in infinite dimensions.* The first half of Corollary 6.8 needs only Proposition 6.2, even on an infinite-dimensional state space. A finite-dimensional energy eigenspace and an isotypic component are both invariant, so their intersection is too. If it is not zero, it contains an irreducible invariant subspace of type $i$, of dimension $n_i$.

*An orthonormal adapted basis exists.* With an invariant inner product (Lecture 4), take the copies mutually orthogonal, choose an orthonormal basis of the first copy, and carry it to the other copies by unitary isomorphisms of representations. Schur's lemma allows the latter: an isomorphism $T$ between unitary irreducibles has $T^\dagger T = c\,\mathrm{id}$ with $c > 0$, so $T/\sqrt{c}$ is unitary.

*Keeping only $C_3$, and reality.* On the triangle of Example 6.3, keep only the rotations $C_3$. Let $\omega = e^{2\pi\mathrm{i}/3}$ and

$$v_k = \sum_{j=1}^{3} \omega^{-kj}\,|j\rangle\,, \qquad k = 0, 1, 2.$$

Since $\omega^{-1} = \omega^2$ and $\omega^{-3} = 1$, these are

$$v_0 = u_0, \qquad v_1 = \omega^2\,|1\rangle + \omega\,|2\rangle + |3\rangle\,, \qquad v_2 = \omega\,|1\rangle + \omega^2\,|2\rangle + |3\rangle\,.$$

The rotation sends $|j\rangle \mapsto |j+1\rangle$, with the labels taken cyclically, $|4\rangle = |1\rangle$, so

$$\rho(r)v_1 = \omega^2\,|2\rangle + \omega\,|3\rangle + |1\rangle = \omega\big(\omega^2\,|1\rangle + \omega\,|2\rangle + |3\rangle\big) = \omega\,v_1,$$

and in the same way $\rho(r)v_k = \omega^k v_k$: the line $\mathbb{C}v_k$ carries $\chi_k$. The $v_k$ are not a guess: they are what the character projectors of $C_3$ give. Since $\rho(r^j)\,|3\rangle = |j\rangle$ and $\chi_k(r^j) = \omega^{kj}$, the projector

$P_k$ of $\chi_k$ gives $P_k\,|3\rangle = \frac{1}{3}\sum_{j=1}^{3}\omega^{-kj}\,|j\rangle = \frac{1}{3}v_k$. Each of the three characters of $C_3$ occurs once, so each $v_k$ is an eigenvector of $H$. Since $1+\omega+\omega^2=0$, the components of $v_1$ and $v_2$ sum to zero: both lie in $V_0$, so their energy is $t$, while that of $v_0 = u_0$ is $-2t$. But $\chi_1$ and $\chi_2$ are different representations of $C_3$, and $C_3$ alone does not force their energies to agree. What does?

The reflections of $C_{3v}$ force it, and so, independently, does the reality of the Hamiltonian. First, the reflections reverse the direction round the triangle and swap the lines of $v_1$ and $v_2$, which together make up $E$. The mirror $s$ of Figure 6.1, for one, swaps $|1\rangle$ and $|2\rangle$ and fixes $|3\rangle$:

$$\rho(s)v_1 = \omega^2\,|2\rangle + \omega\,|1\rangle + |3\rangle = v_2.$$

Since $\rho(s)$ commutes with $H$, if $v_1$ has energy $\lambda$, then so does $v_2$: $Hv_2 = \rho(s)Hv_1 = \lambda\,\rho(s)v_1 = \lambda\,v_2$. Second, the Hamiltonian is *real*: complex conjugation, which for a spinless particle is time reversal, commutes with $H$ and turns $v_1$ into $v_2$,

$$\overline{v_1} = \overline{\omega^2}\,|1\rangle + \overline{\omega}\,|2\rangle + |3\rangle = \omega\,|1\rangle + \omega^2\,|2\rangle + |3\rangle = v_2,$$

so the two have the same energy. The second reason is general. For a Hamiltonian and representation matrices $D(g)$ that are real in one basis, conjugation carries the isotypic component of an irreducible $V^{(i)}$ to that of its conjugate $\overline{V^{(i)}}$. Take as adapted basis of the second the conjugates of the adapted basis of the first. Then the reduced matrices obey $F^{(\bar{\imath})} = \overline{F^{(i)}}$, and since the eigenvalues of $F^{(i)}$ are energies, hence real, the two blocks have the same eigenvalues. So irreducibles with non-real characters come in pairs with equal energies, and the pair forms one real multiplet. A degeneracy that looks accidental for the group at hand is often enforced by reality, or by a larger group.

*Real, complex and quaternionic irreducibles.* In the language of real types (Frobenius–Schur theory, stated without proof): an irreducible is of *real type* if it is equivalent to one by real matrices, of *complex type* if it is not equivalent to its conjugate, and of *quaternionic type* if it is equivalent to its conjugate but has no real form (it cannot be written with real matrices). The three cases are told apart by the Frobenius–Schur indicator

$$\iota_\chi = \frac{1}{|G|}\sum_{g}\chi(g^2) \in \{1, 0, -1\}.$$

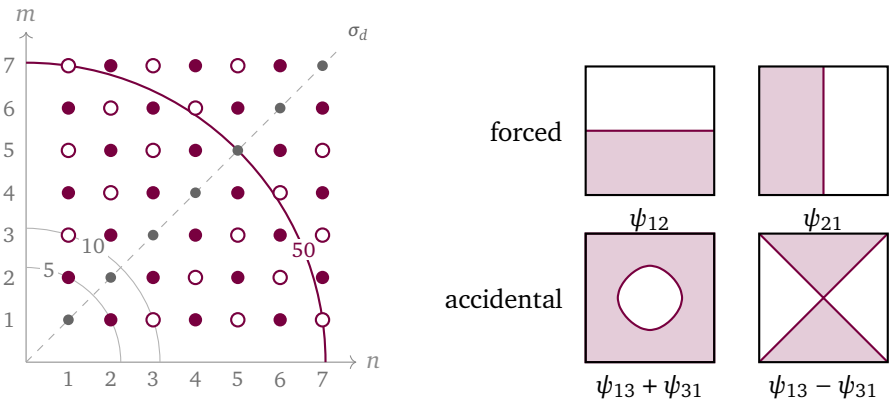
An irreducible of complex type must occur together with its conjugate in a real system, as $\chi_1$ and $\chi_2$ of $C_3$ did on the triangle. The planar rotation representation of $C_3$ shows the same thing for vibrations: a real matrix that commutes with the rotation $R(2\pi/3)$ has the form $a\mathbb{1} + b\,R(\pi/2)$ with $R(\pi/2) = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, the quarter-turn; if it is also symmetric, then $b = 0$, and so the two conjugate complex singlets form a real doublet.

**Example 6.10** (The square well: forced and accidental degeneracy side by side)**.** A particle in a two-dimensional infinite square well $(x, y) \in [-\frac{a}{2}, \frac{a}{2}]^2$ has the symmetry group $D_4$ of the square. The eigenfunctions factorise: $\psi_{nm} = \varphi_n(x)\varphi_m(y)$ with $E_{nm} \propto n^2 + m^2$. Here $\varphi_n$ is a cosine for odd $n$ and a sine for even $n$, so that $\varphi_n(-x) = (-1)^{n+1}\varphi_n(x)$.

Track the $D_4$ action on the pair $W_{nm} = \mathrm{span}\{\psi_{nm}, \psi_{mn}\}$, $n \neq m$. The quarter-turn sends $\psi_{nm} \mapsto (-1)^{m+1}\psi_{mn}$, so its matrix is off-diagonal. The half-turn multiplies by $(-1)^{n+m}$. The axis reflections multiply by $(-1)^{n+1}$ and $(-1)^{m+1}$, and the diagonal reflection swaps the pair. The classes are $(e,\ r^2,\ 2r,\ 2\sigma_v,\ 2\sigma_d)$: $\sigma_v$ are the two axis reflections and $\sigma_d$ the two diagonal ones (the classes of $s$ and $rs$ in Lecture 5's table of $D_4$). The character of $W_{nm}$ is therefore

$$\chi_W = \big(2,\ 2(-1)^{n+m},\ 0,\ (-1)^{n+1} + (-1)^{m+1},\ 0\big).$$

For $n+m$ *odd* this is $(2,-2,0,0,0)=\chi_5$, the two-dimensional irreducible representation of $D_4$. So the degeneracy of, say, $E_{12}$ is *forced* by symmetry. For $n,m$ of *equal* parity the character is $(2,2,0,\pm2,0)$. It decomposes into two *inequivalent one-dimensional* representations: the states $\psi_{nm}\pm\psi_{mn}$ belong to different symmetry types. Their shared energy, for instance $E_{13}=E_{31}$, is an *accidental* degeneracy of this special separable potential. A generic $D_4$-symmetric perturbation of the well will split it. The well even harbours purely arithmetic accidents: $E_{17}=E_{55}$, from $1+49=25+25$, connects different $(n,m)$ pairs altogether. One simple system shows all three phenomena (Figure 6.4).




(a) equal energies share a circle

(b) two degenerate pairs

Figure 6.4: The square well of Example 6.10. (a) The state $\psi_{nm}$ is a lattice point, and $E_{nm}\propto n^2+m^2$, so states of equal energy lie on one circle. Filled dots ($n+m$ odd) form forced pairs, open dots ($n\neq m$ of equal parity) accidental ones, and grey dots on the diagonal have no partner. The circle $n^2+m^2=50$ meets three states, an arithmetic accident. (b) Nodal lines in colour. A quarter-turn carries $\psi_{12}$ into $\psi_{21}$, up to sign: one doublet. The combinations $\psi_{13}\pm\psi_{31}$ share an energy but differ in symmetry ($\chi_1$ and $\chi_3$ in Lecture 5's table of $D_4$).

*Hydrogen.* The levels of the hydrogen atom are $n^2$-fold degenerate (ignoring spin), far more than the $(2\ell+1)$-fold multiplets of the rotation group. The extra degeneracy is enforced by a hidden four-dimensional rotation symmetry, generated by the angular momentum and the Runge–Lenz vector. A starred part of Lecture 13 touches on it.

*Partner functions: aligning the copies.* The character projector $P_i$ picks out the isotypic component but does not see inside it. When the matrices $D^{(i)}$ themselves are known, unitary, the operators

$$P^{(i)}_{\mu\nu}=\frac{n_i}{|G|}\sum_{g\in G}\overline{D^{(i)}_{\mu\nu}(g)}\;\rho(g)$$

resolve $V_i$ direction by direction. On a symmetry-adapted basis of any copy of $V^{(i)}$ they act by $P^{(i)}_{\mu\nu}\,w_{\beta,a}=\delta_{\nu\beta}\,w_{\mu,a}$, and they annihilate the other types; the proof is one line from the great orthogonality theorem of Lecture 5's self-study section. So $P^{(i)}_{\mu\mu}$ projects onto the $\mu$th direction of each type-$i$ copy, and $P^{(i)}_{\mu\nu}$ carries the $\nu$th partner into the $\mu$th. Applied to a few trial vectors, they produce a whole symmetry-adapted basis, partners included, with the copies aligned as Section 6.3 requires.

**Summary.**

- *Quantum symmetry.* $G$ is a symmetry when $[\rho(g),H]=0$ for every $g\in G$; then every

energy eigenspace carries a representation of $G$ (Definition 6.1, Proposition 6.2).

- *Projectors.* $P_i = \frac{n_i}{|G|} \sum_g \overline{\chi_i(g)} \rho(g)$ projects onto the isotypic component $V_i$; $V = \bigoplus_i V_i$, orthogonally when $\rho$ is unitary, and the columns of $P_i$ span $V_i$ (Theorem 6.5).
- *Block diagonalisation.* In an adapted basis, listed partner by partner, $\rho(g) = D^{(i)}(g) \otimes \mathbb{1}$ on $V_i$, and an operator commuting with $G$ acts by $\mathbb{1} \otimes F^{(i)}$: $n_i$ copies of an $m_i \times m_i$ matrix $F^{(i)}$, with $F^{(i)}_{ab} = \langle w_{\nu,a}, A\, w_{\nu,b} \rangle$ in an orthonormal basis (Theorem 6.7). Group theory fixes the blocks, dynamics fills them in.
- *Degeneracy.* $\dim \ker(A - \lambda \,\mathrm{id}_V) = \sum_i n_i \dim \ker(F^{(i)} - \lambda \mathbb{1}_{m_i})$: a level of type $i$ is at least $n_i$-fold degenerate, and generically exactly (Corollary 6.8); more is accidental, or hidden.
- *The triangle.* The projectors give $\mathbb{C}^3 = L \oplus V_0$, of dimensions 1 and 2. In an orthonormal adapted basis the recipe gives $F^{(A_1)} = -2t$ and $F^{(E)} = t$, so $H = \mathrm{diag}(-2t, t, t)$ there: an enforced doublet at $t$. On two stacked triangles each irreducible occurs twice, and two $2 \times 2$ reduced matrices give the singlets $-2t + \beta_\pm$ and the doublets $t + \beta_\pm$.

## References

Tinkham, *Group Theory and Quantum Mechanics*, Ch. 4–5 (symmetry and degeneracy); Serre, *Linear Representations of Finite Groups*, §2.6 (the canonical decomposition); Elliott–Dawber, *Symmetry in Physics*, Ch. 4.